

AMBIKA B

Mechanical Engineer

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ABOUT ME

Mechanical Engineering undergraduate with hands-on experience in CAD design, vehicle subsystem design, product development, reverse engineering, and prototype development. Proficient in SolidWorks, Siemens NX/UG, AutoCAD, Fusion 360, and ANSYS FEA, with experience developing CAD assemblies, fabrication drawings, BOMs, and engineering documentation. Contributed to Team KIOT BAJA's AIR 1 overall championship performance and multiple engineering design competitions.

TECHNICAL SKILLS

CAD Software: SolidWorks, Siemens NX/UG, Fusion 360, AutoCAD
Engineering: Structural Analysis, Product Development, Reverse Engineering, Engineering Graphics, GD&T, Prototype Development, Product Benchmarking, PLM
Documentation: BOM Preparation, Fabrication Drawings, Technical Documentation, MS Word, MS Excel
Programming: Python
Key Roles: Design Team Head (DDR), SAEISS Member, ADMMS Conference Volunteer, MAP Project Head, Placement Ambassador

EXPERIENCE

CAD Designer | ***Team BAJA – E-BAJA & M-BAJA*** *Nov 2025 – Jan 2026*

- Design activities for BAJA vehicle development, supporting prototype design, fabrication, and engineering documentation.
- Designed and assembled 10+ CAD assemblies for off-road vehicle subsystems using Siemens NX and SolidWorks.
- Created 15+ fabrication drawings to support manufacturing and assembly activities.
- Contributed to securing ₹8 lakh in SAE sponsorship funding for prototype development and fabrication.
- Supported vehicle design optimization through frame geometry development and structural considerations.

PROJECTS

Loop Frame & Z-Structure Bicycle Design *2024 – 2025*

- Developed lightweight bicycle frame concepts using Siemens NX and SolidWorks.
- Optimized frame geometry to improve structural stability and rider comfort.
- Integrated a carrier-cum-stand and adjustable seat mechanism into the bicycle design.

Passive Cooling for Home *2024 – 2025*

- Developed a sustainable passive cooling solution using thermal insulation and airflow principles without electrical power.
- Fabricated and validated a functional prototype

IoT-Enabled Intelligent Sewer Monitoring and Flood Risk Early Warning System *2025 – 2026*

- Developed a low-cost IoT drainage monitoring system using ESP8266 and sensors for real-time water level, flow, rainfall, blockage, and overflow detection.
- Integrated Wi-Fi dashboard, alerts, LCD display, and CSV logging for early drainage blockage and flood -risk detection.

Low-Cost Solar Thermal Water Heating System *2025 – 2026*

- Designed a semi-cylindrical solar collector with LDR-based automatic sun tracking to improve solar-energy utilization.
- Created the complete SolidWorks CAD assembly, fabrication drawings, and BOM for prototype development.

Autonomous Medical Supply Delivery Robot *2025 – 2026*

- Designed an autonomous mobile robot concept for transporting medicines and medical supplies within hospitals.
- Developed sensor-based navigation with obstacle detection and avoidance for safe movement.

EDUCATION

B.E. Mechanical Engineering — Knowledge Institute of Technology, Salem

CGPA: 8.7/10 (up to VI-Sem) *2023 – Present*

Higher Secondary (HSC) — Bharathi Kalvi Nilayam Higher Secondary School, Nasiyanur — 80.6% *2022 – 2023*

SSLC (10th) — Bharathi Kalvi Nilayam Higher Secondary School, Nasiyanur — All Pass *2020 – 2021*

ACHIEVEMENTS / PUBLICATION

- Contributed to Team KIOT BAJA – AIR 1 and Overall Championship, BAJA SAEINDIA Southern Championship 2026.
- Secured 1st Place in Maneuverability and 1st Place in Suspension with Team KIOT BAJA; total team cash prize: ₹1.40 lakh.
- Won Best Design Award – ₹10,000 at SAE BDC 6th Edition among 100+ participants.
- Secured 2nd Prize – ₹20,000 for overall performance and bicycle design innovation bicycle pattern design.
- Publication of two papers – Ergonomic loop frame bicycle and Optimizing CNC Machine with AI ,Though Conferences.
- Qualified TN Skill Level 1 – Mechanical Engineering CAD; appeared for TN Skill Level 2 examination conducted by Autodesk.
- Secured 3rd Prize – ₹1,000 at Mr. Professor Technical Event, Nandha Engineering College.